Filipe Nascimento

✉ filipedecn@gmail.com ☎ +55 16 981651696 portfolio: filipecn.dev Brazil

Summary

I have a CS Ph.D. and am passionate about computer graphics, geometry processing, physically based simulation, and problem-solving. My experience includes the design/implementation of fluid solvers, real-time rendering techniques, GPU programming, geometric and spatial data structures, and physically based animation research.

Education

PhD in Computer Science — 2017 - 2024
M.S. in Computer Science — 2013 - 2016
B.S. in Computer Science — 2008 - 2012

Institute of Mathematics and Computer Science (ICMC) -
University of São Paulo (USP), São Carlos, São Paulo, Brazil

Skills & Interests

OpenGL/Vulkan/CUDA/CFD/Houdini's HDK/OpenVDB/OpenUSD
Blender/Gaussian Splatting/M. Learning/OpenFOAM

Programming

C/C++(proficient)/Python/Rust *and some experience with*: MATLAB/R

Languages

Portuguese, English, French (Débutant), Japanese (Beginner)

Professional Experience

RnD Software Engineer at ArqGen — Jan 2026 - Present

Working on the procedural generation/development of geometric solutions for architectural projects.
- Model/Development of optimization methods;
- RnD of geometric algorithms/structures;

Sabbatical Year — 2025

During 2025 I got an offer from Adobe Inc., which unfortunatelly I couldn't accept. On the other hand, I had the privilege to publish and present a paper at SIGGRAPH, share my time with my family and spend time developing personal projects and research.

RnD Software Engineer at Blizzard Entertainment — Jan 2023 - Nov 2024

Worked on C++ 3D tools for cinematics at Blizzard Animation:
- Optimization and extension of the proprietary hair system, including the re-design of data structures, research of interpolation methods, and implementation of visual debugging tools inside Houdini;
- Houdini plugin C++ development (mainly SOPs);

I researched and developed tools for artists. Such tools included applications in animation, modeling, and physical simulations. The last page contains some of the animations I worked on.

Software Engineering Intern at Google Inc. (YouTube) — Jan - Mar 2016

Worked on 360^o video support for YouTube app on Sony's PlayStation 4 VR.
I was responsible for producing the base code that would allow the YouTube app to reproduce 360^o videos on Sony's PlayStation 4 with the Virtual Reality headset. The feature would provide users with an immersive experience when watching such videos. My job was to use computer graphics techniques

(such as texture mapping and meshes) along with Sony's API to develop the project.

Scientific Publications

Digital Animation of Powder-Snow Avalanches 2025
Paper at ACM Siggraph 2025 and ACM Transactions on Graphics
Links: doi pdf video code

Digital 3D Animation of Powder-Snow Avalanches 2025
Paper/Thesis at 38th Conference on Graphics, Patterns and Images (SIBGRAPI 2025)
Links: doi

RBF Liquids: An Adaptive PIC Solver Using RBF-FD 2020
Paper at ACM Siggraph Asia 2020 and ACM Transactions on Graphics
Links: doi pdf video

Approximating implicit curves on plane and surface triangulations with affine arithmetic (AA) 2014
Paper at Computers & Graphics Journal (CAG), Volume 40
Links: doi pdf

Approximating implicit curves on triangulations with AA 2012
Paper at XXV SIBGRAPI Conference on Graphics, Patterns and Images
Links: doi pdf

Research Experience

Digital Animation of Powder Snow Avalanches 2021 - 2024
Applied CFD to solve geophysical equations to simulate avalanches and produce animation volumetric data. Developed in C++, including all geometric processing.
PhD Thesis. Graduate Research supported by FAPESP
My thesis consists of the digital animation of powder-snow avalanches. I used mathematical models from geological research in computer simulations. Computer images were generated from the resulting simulation data. Although my main focus was the entertainment industry, I believe the developments of this research can be applied back to engineering tools and also help other fields.

Adaptive Methods for Fluid Simulation 2017 - 2020
Applied adaptive methods for liquid simulation. Developed in C++, including all adaptive geometric and topological structures.
Graduate Research supported by FAPESP
This was a research in collaboration with my coleagues in my PhD. We explored methods that could adapt a quadtree/octree dynamically in a narrow-band around the liquid interface, providing an adaptive particle sampling for Particle-In-Cell methods. The main contribution was on bringing RBF-FD to the solution, not explored by computer graphics research at the time.

Multimaterial Fluid Simulation for Comp. Graphics Sep 2014 - Feb 2015
Visiting Scholar at University of Waterloo (UW), Canada. Worked with dynamic polygonal surfaces, continuous collision detection, and tet meshes.
Supervisor: Christopher Batty
During this time, I actively participated in developing new ideas for multimaterial fluid simulations. Specifically, the project would handle explicit representation of the surface of liquid bodies of different materials interacting together. We utilized triangular meshes for representing liquid surfaces and interfaces. As multiple liquids interact and move through time, such meshes eventually intersect and get very deformed. The challenge was then to geometrically solve such interactions and many other issues inherent to these types of simulations while keeping the simulation computations robust.

Robust Polygonal Approximation of Implicit Curves and Surfaces 2011-

Undergraduate and Graduate Research supported by FAPESP. 2015
Supervisor: Afonso Paiva

During my undergrad and master's, I participated in research in the Applied Mathematics field. We wanted to achieve the reliable and efficient drawing of implicit curves (sets of roots of mathematical equations). The challenge is that finding the solutions for equations is not trivial in many cases, and searching by brute force algorithms is very costly computationally. My research focused on efficient ways to search for the location of such curves. For doing so, I applied mathematical tools such as Interval Arithmetic and Affine Arithmetics and computational data structures - quadtrees and adaptive triangular meshes.

Activities

Programming Contests (ACM-ICPC)

- World Finals (as coach) – Yekaterinburg, Russia 2014
- Latin America Regional Contest (1st Place) (as coach) 2013
- Latin America Regional Contest 2009
- Brazilian Regional Contest 2009 - 2012

Teaching Assistant

- Advanced Algorithms Laboratory Course 2013
- Introduction to Programming Course 2018 -2019

Other Projects

Over the years I implemented several projects for diverse applications and purposes. Most of them have can be found in the following links:

- https://filipecn.dev/
- https://github.com/filipecn

Applications vary between graphics tools, game engine related tools and applications, GPU algorithms, research paper implementations, machine learning and stock market tools. Combined, they sum up to hundreds of thousands of lines of code, all open sourced.

Most of these projects were created to help me and eventually my colleagues on our daily tasks. For example, in 2022 I was fortunate to help a colleague with Machine Learning code in her research on Parkinson's disease.

Scientific Publications

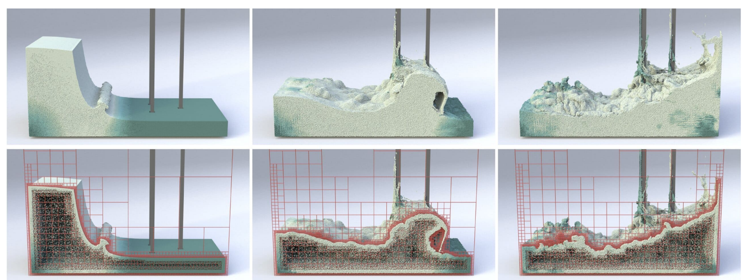
Digital Animation of Powder-Snow Avalanches 2025
Paper at ACM Siggraph 2025 and ACM Transactions on Graphics
Authors: Filipe Nascimento, Fabricio S. Sousa, Afonso Paiva
Links: doi pdf video code

Details: This publication is the direct result of my PhD.
Journal impact factor: 9.5

Digital 3D Animation of Powder-Snow Avalanches 2025
Paper/Thesis at 38th Conference on Graphics, Patterns and Images (SIBGRAPI 2025)
Authors: Filipe Nascimento, Afonso Paiva
Links: doi

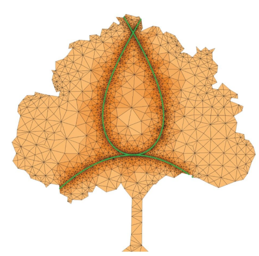
Details: This publication is the comprised version of my thesis.

RBF Liquids: An Adaptive PIC Solver Using RBF-FD 2020
Paper at ACM Siggraph Asia 2020 and ACM Transactions on Graphics
Authors: Rafael Nakanishi, Filipe Nascimento, Rafael Campos, Paulo Pagliosa, Afonso Paiva
Links: doi pdf video



Details: I was the lead programmer of the project, helping to design and implement the whole application. I also contributed with visual tools for debugging and profiling. My main contribution was on the development and implementation of the new data structures we introduced in the paper.
Journal impact factor: 5.414

Approximating implicit curves on plane and surface triangulations with affine arithmetic (AA) 2014
Paper at Computers & Graphics Journal (CAG), Volume 40
Authors: F.C. Nascimento, Afonso Paiva, L.H. Figueiredo, J. Stolfi
Links: doi pdf



Details: This publication is the direct result of my masters.
Journal impact factor: 1.936

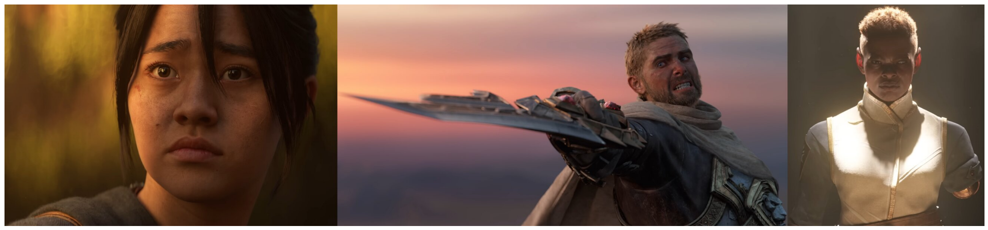
Approximating implicit curves on triangulations with AA 2012
Paper at XXV SIBGRAPI Conference on Graphics, Patterns and Images
Authors: Afonso Paiva, F.C. Nascimento, L.H. Figueiredo, J. Stolfi
Links: doi pdf

Details: I was responsible for the code development and collaborated with writing.

Artistic Work

From my time at Blizzard Entertainment I was fortunate to work on high quality animations with talented artists:

- Diablo IV | Vessel of Hatred (ANNECY 2025 winner in Society for Game Cinematics Distinction category)
- The War Within Announce Cinematic | World of Warcraft
- The War Within Official Cinematic | Shadows Beneath | World of Warcraft



Apart from my research and professional life, I keep artistic hobbies, such as drawing :) :